

Definition Compact operators.

$T: X \rightarrow Y$ is compact if $T(B)$ is precompact in Y .

B : unit ball in X centered at 0. $B = \{x \in X : \|x\| \leq 1\}$

precompact means compact closure.

T compact maps bounded families to compact families

T compact iff for each sequence $(x_n)_{n \in \mathbb{N}} \in X$

with $\|x_n\|_X \leq C$, there is a subsequence $x_{\varphi(n)}$

such that $T x_{\varphi(n)}$ converges in Y .

$\mathcal{B}(X)$ or $\mathcal{B}(Y)$

↓

• X, Y $\mathcal{B}$ -spaces $(\mathcal{B}(X, Y), \|\cdot\|)$ is a Banach space and an algebra.

• Let $\mathcal{K}(X, Y)$ be the subspace of compact operators in $\mathcal{B}(X, Y)$.

Prop • $\mathcal{K}(X, Y)$ is a closed subspace of $\mathcal{B}(X, Y)$.

∴ This means $\lambda T + \mu S$ compact when T, S compact (obvious) but mostly that
 $T_n \rightarrow T$ uniformly and T_n compact $\Rightarrow T$ compact. {quite useful in practice}

• $\dim \text{Ran}(T) < \infty \Rightarrow T$ compact (by Heine-Borel)

• $S \in \mathcal{K}(\cdot, \cdot)$ and $T \in \mathcal{B}(\cdot, \cdot)$ then ST compact, when
 TS compact defined.

• All above straight forward except: T_n compact and $\|T_n - T\| \rightarrow 0 \Rightarrow T$ compact

• T_n compact ∴ T_1 compact $T_1 x_{\varphi_1(m)} \rightarrow y_1 \dots T_j x_{\varphi_j(m)} \rightarrow y_j \dots T_k x_{\varphi_k(m)} \rightarrow y_k$ for all $k \leq j$.

$$\|x_m\| \leq 1$$

$T x_m$ compact?

• $z_j = x_{\varphi_j \circ \dots \circ \varphi_1(j)}$ subsequence with $T_k z_j \rightarrow y_k$ as $j \rightarrow \infty$.

• $T(z_m - z_n) = \underbrace{(T - T_p)(z_m - z_n)}_{\|\cdot\| \leq 2\|T - T_p\| \rightarrow 0} + T_p(z_m - z_n)$. $T_p(z_m - z_n) \rightarrow y_p - y_p = 0, m, n \rightarrow \infty$
So $T z_m$ Cauchy $\Rightarrow$ converges! $\square$

Definition Strong Convergence $T_n \in \mathcal{B}(X, Y)$ converges to T strongly if $\lim_{n \rightarrow \infty} T_n x = Tx$ for all $x \in X$.

This means: $\|T_n x - Tx\|_Y \rightarrow 0$ for all $x \in X$ (in strong topology for Y).

Theorem: $T_n \rightarrow T$ uniformly $\Rightarrow T_n \rightarrow T$ strongly

proof: obvious since $\|T_n x - Tx\|_Y \leq \|T_n - T\| \|x\|_X$.

⚠ $\Leftarrow$ not true. So, convergence in uniform norm is stronger than strong convergence !

• Uniform boundedness theorem (Banach-Steinhaus theorem). ! not in [H.N.] !

• X, Y Banach. $(T_i)_{i \in I} \in \mathcal{B}(X, Y)$ (I index family; not necessarily countable).

Assume $\sup_{i \in I} \|T_i x\|_Y < \infty$ for all $x \in X$. Then there is $C > 0$ s.t.

$$\|T_i x\|_Y \leq C \|x\|_X \text{ for all } x \in X \text{ and } i \in I. \quad \underline{\underline{C: \text{uniform bound}}}$$

• proof uses Baire's theorem as the open mapping theorem.

• Corollary X, Y Banach spaces. $T_n \in \mathcal{B}(X, Y)$ and $T_n \rightarrow T$ strongly.

Then $\sup_n \|T_n\| < \infty$ and $T \in \mathcal{B}(X, Y)$

$$\text{with } \|T\| \leq \liminf_{n \rightarrow \infty} \|T_n\|.$$

Application of strong convergence: Lax equivalence theorem.

A bounded $X \rightarrow Y$ and A_h sequence of approximations $h \equiv \frac{1}{n} \rightarrow 0$

Consider $Au = f$ and $A_h u_h = f_h$. Assume $A_h: X \rightarrow Y$ bijection, A_h^{-1} bounded.

Def. Approximation scheme is convergent if $f_h \rightarrow f \Rightarrow u_h \rightarrow u$.
consistent if $A_h v \xrightarrow{h \rightarrow 0} Av$ for $v \in X$.
stable if $\|A_h^{-1}\| \leq M \quad \forall 0 < h \leq h_0$.

operator form discount

Theorem A consistent scheme is convergent if it is stable.

• proof $\Leftarrow$ (useful direction). $u - u_h = A_h^{-1} [(A_h - A)u + f - f_h]$
so $\|u - u_h\| \leq \|A_h^{-1}\| (\|(A_h - A)u\| + \|f - f_h\|) \rightarrow 0$.

$\Rightarrow u_h = A_h^{-1} f$. $u_h \rightarrow u \Rightarrow A_h^{-1} f$ bounded for all f
for $f_h = f$

Uniform boundedness $\Rightarrow \|A_h^{-1}\| \leq C$ uniformly in h . $\square$

• Applications of convergence to continuous groups.

• $A: X \rightarrow X$ bounded X Banach. Define $e^A := \sum_{k=0}^{\infty} \frac{A^k}{k!}$. $\|e^A\| \leq e^{\|A\|}$
so e^A defined in $\mathcal{B}(X)$.

O.D.E. $\begin{cases} \dot{x} = Ax \\ x(0) = x_0 \end{cases} \Rightarrow x(t) = e^{tA} x_0$. Define $T(t) = e^{tA}$ solution operator.

• $T(t)$ is a uniformly continuous group if :
(i) $T(0) = Id$;
(ii) $T(t+s) = T(t)T(s) \quad \forall s, t \in \mathbb{R}$;
(iii) $T(t) \rightarrow Id$ uniformly as $t \rightarrow 0$

} definition for all

• Check that $T(t) = e^{tA}$ above is uniformly continuous group.

• $T(t)$ is a strongly continuous group when (iii) is replaced by :

(iii') : $T(t) \rightarrow Id$ strongly as $t \rightarrow 0$.

• When A unbounded, this is the best we can hope for.

Application of strongly continuous groups: the shifts. 平移算子

$f(x) \in C_0(\mathbb{R}) = \overline{C_c(\mathbb{R})}$ (continuous functions vanishing at ∞).

$T_h : C_0(\mathbb{R}) \rightarrow C_0(\mathbb{R})$ | bounded
 $f(x) \mapsto T_h f(x) = f(x+h)$ | Linear operator.

$T_0 = I$ • $T_{h_1+h_2} = T_{h_1} T_{h_2}$ group property. • $T_h \xrightarrow{h \rightarrow 0} I$ in what sense?

Answer is $T_h \rightarrow I$ strong but not uniform.
 strong: $Ih f \rightarrow f$ as $h \rightarrow 0$
 uniform: $\|Ih - I\| \rightarrow 0$

$T_h f \xrightarrow{h \rightarrow 0} f$ for $f \in C_0(\mathbb{R})$ i.e. $\|f(x+h) - f(x)\|_\infty = \sup_{x \in \mathbb{R}} |f(x+h) - f(x)| \xrightarrow{\textcircled{*}} 0$ as $h \rightarrow 0$.

This is a statement on uniform continuity of f . But $f \rightarrow 0$ at ∞ . So, $|f(x)| \leq \varepsilon$ for $|x| \geq M$.

And f is uniformly continuous on $|x| \leq 2M$. This implies $\textcircled{*}$ holds (check).

Take $f_h = \begin{cases} 1 & -h \leq x \leq h \\ 0 & \text{else} \end{cases}$. Then $\|T_h f_h - f_h\| = 1 \not\rightarrow 0$ $T_h \not\rightarrow I$ uniformly.

For A bounded, we saw $T(t) = e^{tA}$ uniformly continuous group.
 $\rightarrow A \in B(X)$

For shift $T(h) = T_h = e^{h \frac{\partial}{\partial x}}$ formally by Taylor expansion.

导数算子 $\rightarrow A \equiv \frac{\partial}{\partial x}$ unbounded. 7 $T_h = e^{h \frac{\partial}{\partial x}}$ strongly continuous group.

• Dual Spaces. Coordinates: $x_i: \mathbb{R}^n \rightarrow \mathbb{R}$
 $x \mapsto x_i(x) := \langle \underline{e_i}, x \rangle$ | Generalization to infinite dimensions.

• Def X vector space. The space of continuous linear functionals (maps) from X to $\mathbb{R}$ is called the (topological) dual to X .

• Notation $X^* = \mathcal{B}(X, \mathbb{R}) = \{\varphi: X \rightarrow \mathbb{R} \mid \varphi \text{ cont. linear}\}$

• X nvspace on X^* , we have $\|\varphi\| := \sup_{x \neq 0} \frac{|\varphi(x)|}{\|x\|}$ uniform norm.

• φ bounded implies $|\varphi(x)| \leq \|\varphi\| \|x\|$, $\|\varphi\| < \infty$.

• Notation $\varphi(x) = \langle \varphi, x \rangle_{X^*, X}$ duality product.

• Since $\mathbb{R}$ is Banach, then X^* is automatically Banach.

• In $\mathbb{R}^n$, $\varphi(x) = \varphi\left(\sum_{i=1}^n x_i e_i\right) = \sum_{i=1}^n x_i \varphi(e_i) = \sum_{i=1}^n \varphi_i \langle \underline{e_i}, x \rangle = \sum_{i=1}^n \varphi_i x_i$

so φ linear combination of basis elements $x_i(x)$; $\varphi = \sum_{i=1}^n \varphi_i x_i$.

Thus, since there are n finite coordinates, $(\mathbb{R}^n)^* \stackrel{=}{=} \mathbb{R}^n$.

行，我们把“同构怎么得出”的逻辑压到最核心的 3 步：定义一个映射 $\rightarrow$ 证明它线性 $\rightarrow$ 证明它双射 (= 一一对应)。这就是“ $\cong$ ”的来源。

$\rightarrow$ by Riesz Thm

$(\mathbb{R}^n)^* \cong \mathbb{R}^n$. This identification holds for Hilbert spaces but not for general B. spaces.

$1 < p < \infty$, we have (not an easy result) $L_p^* = L_q$ for $\frac{1}{p} + \frac{1}{q} = 1$.

$p \neq q$ $L_p \not\cong L_q$. $p=2$ $L_2^* \cong L_2$.

$X = C([a,b])$. Let p be smooth function and $\varphi_p(f) := \int_a^b p(x) f(x) dx$.

Then $\varphi_p \in X^*$. $\rightarrow$ Dirac delta (a linear function)

take another $p \rightarrow$. Take $p := \delta_{x_0}(x)$ so that $\langle \delta_{x_0}, f \rangle_{X^*, X} := f(x_0) = \int_a^b \delta_{x_0}(x) f(x) dx$.

So, $\delta_{x_0} \in X^*$ as well.

We can show (not easy) that $X^* = \mathcal{M}_b([a,b])$ space of Radon measures.

This is a Riesz representation theorem, stating that for each $\varphi \in X^*$,

there is $\mu \in \mathcal{M}_b([a,b])$ s.t. $\varphi(f) = \int_a^b f d\mu \equiv \int_a^b f \mu(dx)$.

Another Riesz representation states that $\forall \varphi \in L_p^*(X)$, there is $\psi \in L_q(X)$

s.t. $\varphi(f) = \langle \psi, f \rangle_{L_q, L_p} = \int_X \psi(x) f(x) dx$,

在 $C([a,b])$: 对偶 = 测度

在 L^p : 对偶 = L^q 函数

• Hahn-Banach theorem.

• Assume a linear functional constructed on a subspace of X . We want to extend the functional to X without increasing its norm.

• Theorem . Let subspace $Y \subset X$ n.v.sp. and $\varphi: Y \rightarrow \mathbb{R}$ bounded linear functional with $\|\varphi\|_{Y^*} = M < \infty$.

• Then there is $\varphi: X \rightarrow \mathbb{R}$ bounded linear, such that

$$\varphi|_Y = \varphi \quad \text{and} \quad \|\varphi\|_{X^*} = \|\varphi\|_{Y^*}.$$

• More general versions replace the norm by more general sub-additive functions.

• The proof is based on the axiom of choice or equivalently Zorn's lemma.

• Let $y_1, y_2 \in Y$; $x \notin Y$. $\tilde{\varphi}(y_1) - \tilde{\varphi}(y_2) = \tilde{\varphi}(y_1 - y_2) \leq \|y_1 - y_2\| \leq \|y_1 + x\| + \|y_2 + x\|$.

Define $\tilde{\varphi} = M^{-1}\varphi$ so, $a := \sup_{y_2 \in M} -\tilde{\varphi}(y_2) - \|y_2 + x\| \leq \inf_{y_1 \in M} -\tilde{\varphi}(y_1) + \|y_1 + x\| =: b$

Define $\tilde{\varphi}(y + rx) := \tilde{\varphi}(y) + r$ for $a \leq r = \tilde{\varphi}(x) \leq b$, $r \in \mathbb{R}$.

• above shows $-\|y + rx\| \leq \tilde{\varphi}(y + rx) \leq \|y + rx\|$: take $y = ry_1$ and $a \leq r \leq b$.

• Extend to next " x ". Zorn's Lemma allows one to show $\tilde{\varphi}$ is defined everywhere $\square$

• There are geometric versions of H.B. allowing one to separate disjoint convex subsets by a hyperplane.

• H.B. allows one to show that the dual space X^* is "very informative" about X .

• Corollary $\forall x_0 \in X$, there is $f_0 \in X^*$ s.t. $\|f_0\|_{X^*} = \|x_0\|_X$ and $\langle f_0, x_0 \rangle_{X^*, X} = \|x_0\|_X^2$.

• proof $Y = \mathbb{R}x_0 \subset X$ $\psi(t x_0) = t \|x_0\|^2 \Rightarrow \psi \in Y^*$. By H.B., f_0 extends ψ to X^*

$$\text{with } \|f_0\|_{X^*} = \|\psi\|_{Y^*} = \sup_{t \in \mathbb{R}} \frac{|\psi(t x_0)|}{\|t x_0\|} = \|x_0\|. \quad \square$$

• Corollary (Characterization of norm). Let $x \in X$. Then $\|x\| = \max_{f \in X^*, \|f\|=1} |\langle f, x \rangle|$.

• proof. For $x = x_0$ above, choose $f_1 = \frac{1}{\|x\|} f_0$. Then $\frac{1}{\|x\|} \langle f_0, x \rangle = \|x\|$. Check rest. $\square$

• Corollary (X^* separates X). Let $x, y \in X$. Assume $\varphi(x) = \varphi(y) \forall \varphi \in X^*$. Then $x = y$.

• proof. Take x, y in X . $x - y \in X$ and $\|x - y\| = \langle f, x - y \rangle$, f as above.

$$\text{So } \|x - y\| = \langle f, x \rangle - \langle f, y \rangle = 0.$$

- X^* is a B. space as space of bounded linear operators to $\mathbb{R}$, complete.
- This holds even if X is not a B. space. Then $X^{**} = (X^*)^*$ also B. space.
- Let $x \in X$. Define $F_x(\varphi) = \varphi(x) \quad \forall \varphi \in X^*$. $F_x \in X^{**}$. so $X \subset X^{**}$.
- When all $F \in X^{**}$ can be written as $F(\varphi) = \varphi(x)$ for $x \in X$,
then $X^{**} \subset X$ and $X \cong X^{**}$. X is called reflexive.
- For $1 < p < \infty$, $(L^p)^* = L^q$ $\frac{1}{p} + \frac{1}{q} = 1$ so $(L^p)^{**} = L^p$ reflexive.
- $(L^1)^* = L^\infty$. However, $(L^\infty)^* \neq L^1$. L^1, L^∞ not reflexive.

Notions of weak and weak* convergence.

- Def $x_n \in X$ c.v. weakly to ~~x~~ if $\varphi(x_n) \rightarrow \varphi(x) \quad \forall \varphi \in X^*$.

Notation $x_n \xrightarrow{n \rightarrow \infty} x$.

- strong c.v. $\Rightarrow$ weak c.v. since $|\varphi(x) - \varphi(x_n)| \leq \|\varphi\| \|x - x_n\| \rightarrow 0$.
- weak c.v. $\not\Rightarrow$ strong c.v. in general although it does in finite dimensions.
- Def $\varphi_n \in X^*$ converges weak* to φ if $\varphi_n(x) \rightarrow \varphi(x) \quad \forall x \in X$.

Notation $\varphi_n \xrightarrow{*} \varphi$ as $n \rightarrow \infty$.

- Note: this is the notion of strong convergence for operators (since $\varphi: X \rightarrow \mathbb{R}$).
- For reflexive spaces, these two notions are identical $X \leftarrow X^* \rightarrow X^{**} = X$.
- On X , strong topology generated by $B_\varepsilon(x)$ open; this is $(X, \|\cdot\|)$.
- weak topology $(X, \mathcal{C}_w)$ generated by sets $\varphi^{-1}(\text{open})$ for each $\varphi \in X^*$.
- On X^* , weak* topology $(X^*, \mathcal{C}_{w*})$ generated by $x^{-1}(\text{open})$ for each $x \in X$.
- $(X^*, \mathcal{C}_{w*})$ has less open sets than $(X^*, \mathcal{C}_w)$; less open sets than $(X^*, \|\cdot\|)$.
- Less open sets means : 13 easier to be compact!

- Theorem (Banach-Alaoglu). The closed unit ball in X^* is w^* -compact.
- proof not elementary.
- Result as general as Heine-Borel but for weak* convergence.
- A bounded sequence in $(X^*, \|\cdot\|)$ admits a weak* converging subsequence.
(the weak* topology is not metrizable on X^* ; however, it is on the closed unit ball).
- This is an important existence result in many PDE theories.
- Application. $f_n \in L^2(0,1)$, $\|f_n\| \leq 1$. $L^2 = (L^2)^*$ so weak $\Leftrightarrow$ weak*.

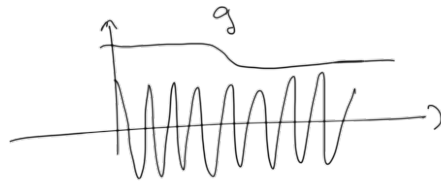
Then $f_{\varphi(n)} \rightarrow f$ in L^2 , which means:

$$\forall g \in L^2(0,1) \quad \int_0^1 f_{\varphi(n)}(x) \bar{g}(x) dx \rightarrow \int_0^1 f(x) \bar{g}(x) dx.$$

- Take $f_n(x) = \cos(nx)$. $f_n \rightarrow f$. Verify $f = 0$

$$\cos nx \rightarrow 0 \text{ in } L^2 \quad \text{but} \quad \int_0^1 \cos^2 nx dx \simeq \frac{1}{2} \text{ so } \cos nx \not\rightarrow 0 \text{ in } L^2(0,1).$$

• $\cos nx \rightarrow 0$



• oscillations of $\cos nx \Rightarrow$ for g piece-wise constant, $\int g \cos nx \rightarrow 0$

• any function in $L^2(0,1)$ can be approximated by piece-wise constant functions.

• In general, weak c.v. in absence of strong c.v. is caused by oscillations.
(Young measures.)

• Theorem (Kakutani). X reflexive ($X = X^{**}$) if and only if

$$B_X = \{x \in X, \|x\| \leq 1\} \text{ is compact for } (X, \tau_w)$$

• This implies that for compactness purposes, weak* is the important notion, and $\text{weak} \Leftrightarrow \text{weak}^*$ for reflexive spaces.

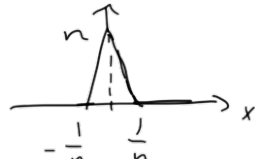
• Example of weak * c.v. $L^\infty = (L^1)^*$; L^∞ is a dual space.

• So $f_n \in L^\infty$, $\|f_n\|_\infty \leq 1 \Rightarrow f_n \xrightarrow{w*} f$ in L^∞ i.e.,

$$\int f_n(x) g(x) dx \rightarrow \int f(x) g(x) dx \quad \forall g \in L^1.$$

• Same for $L^p = (L^q)^*$; $1 < p < \infty$

• However, L^1 is not the dual of any space. It is in fact "too small".

• Assume $\|f_n\|_{L^1} \leq 1$. Take: $f_n(x) = \begin{array}{c} \uparrow n \\ \text{triangle} \\ \downarrow \frac{1}{n} \end{array}$ 

Then $\|f_n\| = 1$. But $f_n \longrightarrow \delta_0(x)$ delta-"function" not in L^1 .

So, f_n converges to an object in $\mathcal{M}_b$: Radon measures

• We know $L^\infty \supset \mathcal{C}$ continuous functions. $\mathcal{C}^* = \mathcal{M}_b \supset L^1$.

• Now $\mathcal{M}_b$ is a dual space. $\|f_n\|_{L^1} \leq 1 \Rightarrow \|f_n\|_{\mathcal{M}_b} \leq 1$ (in appropriate sense.)

• Banach Alaoglu $\Rightarrow f_n \xrightarrow{*} f$ in $\mathcal{M}_b$!

So $\int f_n(x) \varphi(x) dx \rightarrow \int f(x) \varphi(x) dx$ for φ continuous
 (not only bounded).